Senior Health Workforce Analyst - Assignment

Health Workforce Modelling

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Outline

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About me

- Quantitative modeler with 10+ years of experience:
  - Decision models (optimization)
  - Agent-based simulation
  - Physics-based modeling
  - Macroeconomic models
- Currently, Data Analytics Specialist @ MCCSS.
- Goals:
  - Apply data analytics and modeling to inform decision-making and uncover actionable insights.
  - Learn and grow as a professional by taking on new challenges and responsibilities.

Assignment & Modelling Approach

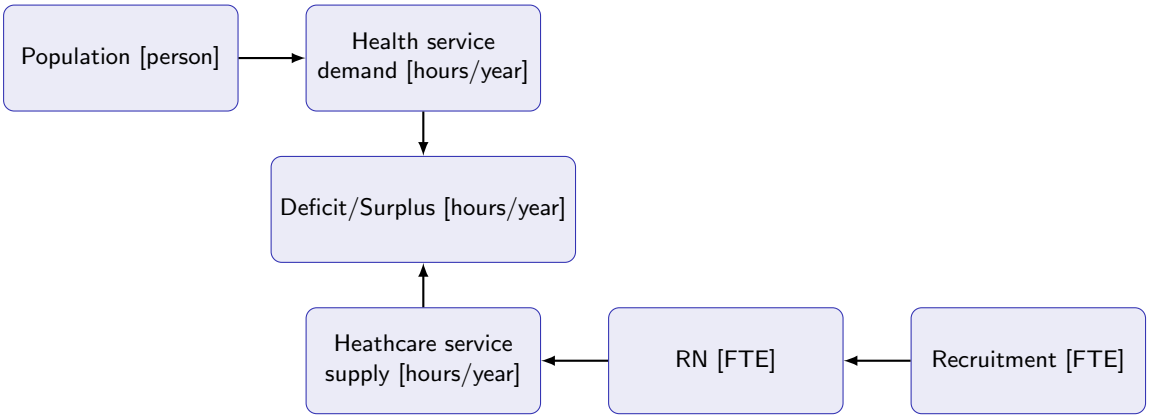
Develop a quantitative model to forecast workforce supply and workforce requirements for **Registered Nurses** (RNs) in Ontario over the next 10 years.

Modelling approach:

- Start simple and enrich only if needed.
- Research or policy question being addressed.
- Decision variables or policy instruments.
- Causal relationships between variables.

There is no good solution to a badly posed problem.

Conceptual formulation



- FTE: Full-time equivalent.

Model Decisions

- Which variables are considered exogenous?
- Spatial and temporal granularity.
- Most appropriate model type:
  - System dynamics
  - Agent-based modeling
  - Optimization
  - Time series
- How to deal with data limitations:
  - Literature coefficients
  - Imputation (e.g., uniform, proportional allocation)
  - Determine assumptions (e.g., visits uniform over the year)
- How to deal with uncertainty:
  - Sensitivity analysis
  - Scenario analysis
  - Probabilistic modeling
- Resource constraints
- Model validation

Conceptualizing supply, requirements and distribution

Supply

- Stock-and-flow, not a registrant headcount
- Registered → active → practising in Ontario
- Converted to FTE (full/part-time, hours)

Requirements

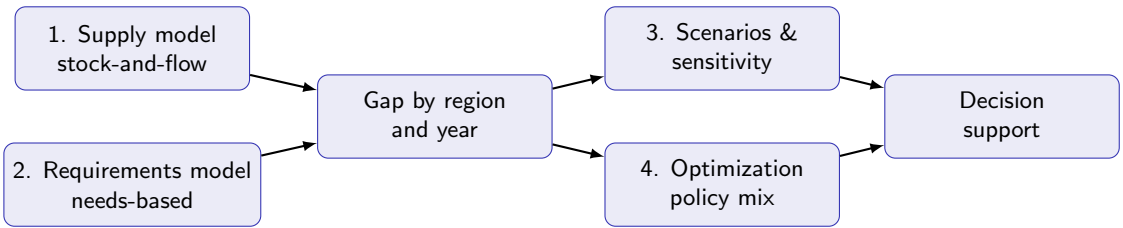
- Needs-based: workforce required to meet projected population and service needs
- Driven by age/sex, population growth, care setting, models of care
- Not the same as current employment or vacancies

Geography

- Modelled by Ontario Health region
- Urban / rural / northern lens
- Province-wide balance can hide regional shortages

$$\textbf{Gap}_{r,t} = \text{Requirement}_{r,t} - \text{Supply}_{r,t} \quad \text{(region } r\text{, year } t\text{, in FTEs)}$$

Modelling pipeline



- Annual time steps, ten-year horizon, a handful of regions: transparent and fast to run.
- Moves from a *descriptive forecast* to *decision support*.

Supply: stock-and-flow (cohort projection)

$$S_{r,a}(t+1) = S_{r,a}(t) + E_r(t) - A_{r,a}(t) + M_{r,a}(t)$$

- **Inflows**
  - New graduates E, lagged by program length and exam pass rates; IEN licensure
  - Net interprovincial / international migration M
  - Re-entrants
- **Outflows**
  - Age- and career-stage-specific retirement / exit hazard rates A (not one flat attrition rate)

- **Key assumptions**
  - Age-specific rates stay at recent levels (baseline)
  - Graduates and IENs stay in Ontario at observed rates
  - FTE per active RN by age and sex
  - Regional distribution of new entrants follows observed patterns

Requirements: needs-based model

$$R_r(t) = \sum_{a,s,c} P_{r,a,s}(t) \times U_{a,s,c} \times I_c(t)$$

- P: population by region, age band and sex (Ontario Ministry of Finance projections)
- U: service utilization rate by age/sex and care setting c (hospital, long-term care, home and community, primary care)
- I: service-intensity adjustment converting service use to RN FTEs, including policy-driven changes (more home care, skill-mix shifts)
- **Cross-check:** ratio benchmarking (RNs per 1,000 population) as a sensitivity lens on the needs-based estimate
- **Key assumptions:** utilization by age stays constant in the baseline; changes in care models enter only as explicit scenarios

Optimization layer: from forecast to recommendation

Decision variables x_t: domestic training seats, IEN licences fast-tracked, retention-incentive spend, inter-regional deployment incentives.

Formulation A: least cost

$$\min \sum_{k,r,t} c_k \, x_{k,r,t}$$

s.t. $S_{r,t}(x) \geq R_{r,t}$ for all r, t

Formulation B: fixed budget

$$\min \max_{r,t} \left(R_{r,t} - S_{r,t}(x)\right)^+$$

s.t. $\sum c_k \, x_{k,r,t} \leq B$

- **Constraints:** training and clinical-placement capacity, annual budget ceilings, realistic ramp-up rates, minimum service level per region.
- Linear / mixed-integer program, open-source solvers (Python with PuLP or OR-Tools): fast and explainable.
- The two formulations are two views of the same cost–gap trade-off.

Optimal hiring policy model

Decision variable x_t: new nurse hired at time t.

$$\min_{x_t} \sum_{t=0}^{T} e^{-\lambda t}\, y_t^+ + \sum_{t=0}^{T} e^{-\mu t}\, y_t^-$$

s.t.

$$d_t = \alpha_t\ p_t \quad \forall t \tag{1}$$

$$s_t = \beta_t\ n_t \quad \forall t \tag{2}$$

$$d_t - s_t = y_t^+ - y_t^- \quad \forall t \tag{3}$$

$$n_t = \gamma\ n_{t-1} + x_t \quad \forall t \tag{4}$$

$$x_t, y_t^+, y_t^- \geq 0 \quad \forall t \tag{5}$$

- **Constraints:** training and clinical-placement capacity, annual budget ceilings, realistic ramp-up

Data sources, variables and indicators

	Sources	**Variables / indicators**
Supply	College of Nurses of Ontario (CNO) registration statistics; CIHI Nursing in Canada / National Nursing Database; program enrolment and graduation counts; licensing exam results	Headcount and FTE; age and sex; region; practice setting; employment status (full-/part-time); graduation, pass and licensing rates; migration
Requirements	Ontario Ministry of Finance population projections; CIHI health service utilization; Ontario Health regional reports; Statistics Canada	Population by region/age/sex; service use by setting; RNs per 1,000 population; hours of care per service unit
Policy	Training-seat funding, IEN pathway capacity, program costs	Unit cost per seat, IEN licence, retention incentive; capacity limits; ramp-up rates

Data quality, gaps and privacy

Quality and gaps

- Registration overstates active, practising supply → adjust using employment status and FTE conversion
- Small regional cells are suppressed or unstable → pool years or regions
- 1–2 year publication lag → nowcast the latest year and flag it
- Self-reported hours are noisy → validate against multiple sources
- Reconcile definitions across CNO, CIHI and StatCan; document every assumption

Privacy

- Aggregate, de-identified data only
- No individual licensing or patient records needed
- Respect CIHI/CNO data-use terms and small-cell suppression rules
- Follow FIPPA principles for any internal administrative data

Assessing uncertainty: scenarios and sensitivity

Scenarios

- **Baseline:** status-quo trends
- **High attrition:** accelerated retirements
- **Expanded seats:** policy intervention
- **High IEN:** fast-tracked licensing / immigration

Sensitivity analysis

- One-at-a-time variation of attrition, utilization growth, completion/pass rates and migration $\rightarrow$ tornado chart of which assumptions move the 2036 gap most
- Monte Carlo draws from plausible parameter ranges $\rightarrow$ uncertainty band around supply and requirements

Robust recommendations

Re-solve the optimization under each scenario: show how the policy mix and its cost change, and favour options that perform well across scenarios.

Communicating results to senior management

1. **Headline chart:** projected supply vs. requirements, 2026–2036, province-wide and by region, with a shaded uncertainty band.
2. **Regional gap view:** when and where gaps first appear.
3. **Optimization result:** the lowest-cost policy mix (seats, IEN, retention, by year) as a stacked bar chart.
4. **Tornado chart:** which assumptions matter most, and so where better data is worth the effort.

- Plain language, one message per slide, assumptions stated up front.
- Brief for executives; technical appendix for analysts; reproducible code.

Limitations and policy implications

Limitations

- Quality depends on data lag and assumption validity
- Does not capture non-linear shocks (e.g. pandemics)
- Regional model is coarser than facility-level planning
- Requirements reflect current care models unless scenarios change them

Policy options (with trade-offs)

- Expand training seats: durable but slow
- Fast-track IEN licensing: quick, limited by pathway capacity
- Retention incentives: fast, recurring cost
- Regional / rural incentives: target maldistribution
- Scope-of-practice / skill-mix change: shifts requirements

Each option is presented with cost, lead time and feasibility to inform government action.

Summary

- A linked **supply** and **needs-based requirements** model, by region, over ten years.
- An **optimization layer** turns gaps into a least-cost policy mix.
- **Scenarios and sensitivity analysis** make recommendations robust.
- Built on public, aggregate data, with explicit quality and privacy safeguards.
- Communicated as options with trade-offs, refreshed annually as new data arrive.

Thank you — questions?